Notes_25.08: Naive Bayes Implementation in Python







🌐 **Language: English** | हिंदी / Hinglish | ⬅️ Lecture 25 Overview

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📌 Executive Summary

Building GaussianNB classifier on real cardiovascular data and evaluating classification performance.

🎯 Mathematical Formulations

Model Log-Likelihood Decision Boundary:

$$\log P(y \mid \mathbf{x}) \propto \log P(y) + \sum_{j=1}^{d} \log P(x_j \mid y)$$

💡 Conceptual Principles & Architecture

Log-Probabilities Kyu Use Karte Hain?

Jab d-dimensions bohot bada hota hai (e.g. text classification mein $10,000$ words), toh small probabilities ($0.001 times 0.002 times \ldots$) multiply karte-karte computer ke floating point precision se bahar nikal kar underflow (0.0) ho jaati hain. Logarithm lene se multiplication addition ban jata hai: $\log(a \cdot b) = \log a + \log b$, jisse underflow bilkul prevent ho jata hai!

💻 Python Production Implementation

```
import pandas as pd
from sklearn.model_selection import train_test_split
from sklearn.naive_bayes import GaussianNB
from sklearn.metrics import classification_report

df = pd.read_csv("heart.csv")
X = df.drop("target", axis=1)
y = df["target"]

X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.2, random_s
gnb = GaussianNB()
gnb.fit(X_train, y_train)
y_pred = gnb.predict(X_test)

print("Classification Report:\n", classification_report(y_test, y_pred))
```